Grade 9 Biology

Solved Exercises - English Medium - Balochistan Textbook Board (BTBB)

Complete book: all 8 chapters, all exercises (multiple choice, short response and extended response questions).

Chapter 1: Nature of Science in Biology

Chapter 2: Molecular Biology

Chapter 3: Cell and Subcellular Organelles

Chapter 4: Tissues, Organs and Systems

Chapter 5: Metabolism

Chapter 6: Biodiversity and Classification

Chapter 7: Form and Function in Plants

Chapter 8: Evolution and Natural Selection

Source: BTBB Grade 9 Biology textbook (National Curriculum 2022-23), 172 pages. Medium confirmed from the book itself: cover, contents and all chapters are in English (the book is a scanned-image PDF with no text layer, so its exercises were read from rendered pages). Questions are reproduced from each chapter exercise; answers are written in medium length.

Chapter 1: Nature of Science in Biology

Exercise 1

A. Multiple choice questions. (Correct option marked with a green tick.)

(1) A biologist is searching new species of plants in an area, the branch of biology relevant to his work is:

(a) Microbiology

(b) Zoology

(c) Botany ✓

(d) Genetics

(2) Metabolism, enzyme actions etc., in a living body are studied in:

(a) Morphology

(b) Chemistry

(c) Cytology

(d) Biochemistry ✓

(3) The study of basic units of a living body is called:

(a) Morphology

(b) Histology

(c) Pathology

(d) Cytology ✓

(4) When we observe different organs inside a human body, it means we are studying:

(a) Morphology

(b) Anatomy ✓

(c) Cytology

(d) Histology

(5) How characters are transferred from parents to their offsprings is studied in:

(a) Histology

(b) Embryology

(c) Genetics ✓

(d) Morphology

(6) The science of growing gardens, fruits and ornamental plants is known as:

(a) Forestry

(b) Farming

(c) Agriculture

(d) Horticulture ✓

(7) The statement that describes the primary focus of ecology is:

(a) The study of individuals, organisms and their characteristics

(b) The study of interactions between organisms and their environment ✓

(c) The study of evolution of species over time

(d) The study of structure and functions of ecosystems

(8) Which of the following statements is accurate about microbiology?

(a) The study of structure and function of cells

(b) The study of evolution of species over time

(c) The study of bacteria, viruses, fungi ✓

(d) The study of interactions between organisms and their environment

(9) A doctor carried out diagnostic tests of a patient to know the actual cause of disease. The branch of biology concerned with these investigations is:

(a) Pharmacology

(b) Immunology

(c) Pathology ✓

(d) Molecular biology

(10) Encircle the wrong statement:

(a) Theory is formulated after experiments

(b) Hypothesis is formulated after observations

(c) Deduction is formulated after experiments ✓

(d) Deduction is formulated from untested hypothesis

B. Short response questions.

(i) How does the knowledge of statistics and economics help in biological studies?

Statistics (biostatistics) helps biologists collect, analyse and judge data from experiments and surveys, for example whether a drug really works. Economics (bioeconomics) helps judge the value of living resources, such as fisheries or forests, and plan their sustainable use.

(ii) How will you differentiate quantitative and qualitative observations?

Quantitative observations are measured in numbers, for example a plant grew 5 cm or a temperature of 37 C. Qualitative observations describe qualities without numbers, for example a leaf is green, rough or sweet-smelling. Quantitative observations can be compared exactly; qualitative observations describe what is seen, heard or felt.

(iii) Briefly describe horticulture, animal husbandry and fisheries.

Horticulture is the science of growing gardens, fruits, vegetables and ornamental plants. Animal husbandry is the care, breeding and management of farm animals for milk, meat, eggs and work. Fisheries is the rearing and catching of fish and other aquatic animals for food.

(iv) Write a brief note on the relation of biology with geography, technology and

physics.

Biology relates to geography through biogeography, the study of where organisms live and why, which depends on climate and land. It relates to technology through instruments such as microscopes, and to physics through biophysics, which uses physical principles to study how living systems work, for example how the eye forms images.

(v) You know that plasmodia are the cause of malaria but a healthy person having plasmodia in the blood does not show symptoms of malaria. Why?

Such a person may carry only a very small number of parasites, or may have developed partial immunity from earlier exposure, so the parasites are kept in check and do not destroy enough red blood cells to cause fever and other symptoms. The person can still pass the parasite on through a mosquito bite.

(vi) Define the following terms: (i) Microbiology (ii) Physiology (iii) Embryology (iv) Histology (v) Palaeontology (vi) Pharmacology (vii) Horticulture (viii) Immunology (ix) Biological method (x) Qualitative observations (xi) Deduction (xii) Scientific theory.

(i) Microbiology: study of microorganisms such as bacteria and viruses. (ii) Physiology: study of the functions of parts of living organisms. (iii) Embryology: study of the development of embryos. (iv) Histology: study of tissues. (v) Palaeontology: scientific study of fossils. (vi) Pharmacology: study of drugs and their effects on cells, tissues and organs. (vii) Horticulture: science of growing gardens, fruits and ornamental plants. (viii) Immunology: study of the immune system and how the body defends itself. (ix) Biological method: the step-by-step scientific method used to solve biological problems. (x) Qualitative observations: observations described in words, without numbers. (xi) Deduction: a logical conclusion drawn from a hypothesis, stating what must happen if the hypothesis is true. (xii) Scientific theory: a well-tested explanation, supported by experiments, that explains a set of observations.

C. Extended response questions.

(i) Relate the verses of the Holy Quran which stress upon the study of biology.

The Holy Quran repeatedly invites people to observe living things and reflect on their creation: the origin of life from water, the stages of human development, the diversity of plants and animals, and the signs in nature for people who think. These verses present the study of life as a way to recognise the Creator, which is why biology begins with reflection on living organisms.

(ii) Explain four steps (observations, hypothesis formulation, deduction and experimentation) of a biological method to solve a biological problem.

Step 1, Observation: notice a problem, for example many people in an area suffer from malaria. Step 2, Hypothesis: propose a testable explanation, for example malaria is spread by mosquitoes. Step 3, Deduction: draw a logical consequence, for example if mosquitoes

spread malaria, people protected from mosquitoes should fall ill less often. Step 4, Experimentation: test the deduction with controlled experiments and record the results, then accept, reject or revise the hypothesis.

(iii) Justify that biology is a part of Natural science.

Natural science studies nature through observation and experiment, and biology does exactly this for living things: it observes organisms, tests ideas by experiments, and builds explanations from evidence. Its branches, such as botany, zoology and ecology, all study the natural world, so biology is a natural science, and also a life science.

(iv) Do you think that science is a collaborative field? Justify.

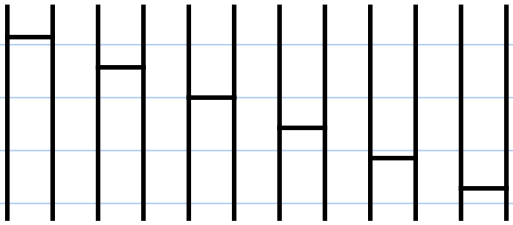
Yes, science is collaborative because no single person can master every field or repeat every experiment. Biologists work with chemists, physicists, doctors, statisticians and technologists, share results through journals, and check one another's work. This sharing and criticism is what makes scientific knowledge reliable and lets it grow.

(v) Malaria in human beings is spread by female anopheles mosquito. Explain it using biological method.

Observation: malaria is common where mosquitoes are common, and patients carry Plasmodium in their blood. Hypothesis: the female Anopheles mosquito carries Plasmodium from an infected person to a healthy person. Deduction: if this is true, the parasite should be found in the mosquito's gut and salivary glands, and people protected from bites should escape infection. Experiment: mosquitoes fed on malaria patients are examined, and bite-protection trials are run. The results confirm the hypothesis, which becomes an accepted explanation.

Chapter 2: Molecular Biology

Diagram: DNA double helix (Unit 2)



Two strands, bases paired: A-T, G-C

Exercise 2

A. Multiple choice questions. (Correct option marked with a green tick.)

(1) One of the following groups contains all polysaccharides:

(a) Sucrose, glucose and fructose

(b) Maltose, lactose and fructose

(c) Glycogen, sucrose and maltose

(d) Glycogen, cellulose and starch ✓

(2) The most abundant biomolecules in living bodies are:

(a) Lipids

(b) Proteins ✓

(c) Carbohydrates

(d) Nucleic acids

(3) Sucrose is formed by the combination of:

(a) Two glucose

(b) Glucose and Fructose ✓

(c) Two fructose

(d) Maltose and Manose

(4) The carbohydrate abundant in plants is:

(a) Glucose

(b) Cellulose ✓

(c) Maltose

(d) Sucrose

(5) The diameter of DNA molecule is about:

(a) 2 A

(b) 4 A

(c) 20 A ✓

(d) 40 A

(6) Proteins are synthesized from:

(a) Glucose

(b) Fatty acids

(c) Amino acids ✓

(d) Nucleotide

(7) Which one is not true about lipids?

(a) Vitamin A and D are lipids in nature

(b) Lipids are reserve energy in the body

(c) They are the main source of energy in living organisms ✓

(d) Prostaglandins are lipids in nature

(8) The pyrimidine nitrogenous bases in RNA are:

(a) Cytosine and uracil ✓

(b) Adenine and Guanine

(c) Thiamine and cytosine

(d) Thiamine and Uracil

(9) The primary function of carbohydrates in living organisms is to:

(a) Provide structural support

(b) Serve as primary source of energy ✓

(c) Store genetic information

(d) Catalyze biochemical reactions

(10) The primary difference between DNA and RNA in terms of structure is:

(a) DNA is single stranded and RNA is double stranded

(b) DNA contains the base uracil while RNA contains thymine

(c) DNA has 2-OH groups while RNA does not

(d) DNA is double stranded while RNA is typically single stranded ✓

(11) The term for the process by which genetic information is passed from DNA to RNA is:

(a) Translation

(b) Transcription ✓

(c) Replication

(d) Mutation

(12) Which of the following is responsible for converting information in DNA to form protein?

(a) Messenger RNA (mRNA) ✓

(b) Transfer RNA (tRNA)

(c) Ribosomal RNA (rRNA)

(d) Small nuclear RNA (SnRNA)

B. Short response questions.

(i) Name the four common biomolecules along with their main roles.

The four common biomolecules are: carbohydrates, the main source of energy and

structural material in plants; lipids, stored energy, insulation and cell membranes; proteins, enzymes, structure, transport and defence; and nucleic acids (DNA and RNA), storage and transfer of genetic information.

(ii) Write a note on the general structure of amino acids.

An amino acid has a central carbon atom joined to four groups: an amino group (-NH2), a carboxyl group (-COOH), a hydrogen atom, and a variable side chain (R group). The R group differs in each amino acid and gives it its special character; amino acids join by peptide bonds to form proteins.

(iii) What are monosaccharides?

Monosaccharides are the simplest sugars, made of a single sugar unit that cannot be broken down into smaller sugars by hydrolysis. Examples are glucose, fructose and galactose; they are sweet, soluble in water and used directly as fuel by cells.

(iv) Write functions of lipids.

Lipids store energy for long-term use, insulate the body against heat loss, protect organs, form cell membranes, and act as hormones and vitamins (such as vitamin A and D). They also carry fat-soluble vitamins in food.

(v) What do you understand by the term glycosidic linkage?

A glycosidic linkage is the chemical bond that joins two sugar units together, formed when a water molecule is removed (condensation) between them. For example, glucose and fructose join by a glycosidic bond to form sucrose.

(vi) What are the hydrolysis products of sucrose and maltose?

Hydrolysis of sucrose gives glucose and fructose. Hydrolysis of maltose gives two molecules of glucose.

(vii) Give a comparison of DNA and RNA.

DNA is double-stranded, contains deoxyribose sugar and the base thymine, and stores genetic information in the nucleus. RNA is usually single-stranded, contains ribose sugar and the base uracil instead of thymine, and carries out the instructions of DNA in making proteins.

(viii) Write down the names of nucleotides found in RNA and DNA.

DNA nucleotides are built from adenine, guanine, cytosine and thymine. RNA nucleotides are built from adenine, guanine, cytosine and uracil.

(ix) Write down the functions of DNA.

DNA stores the genetic information of an organism, passes it to the next generation, directs the making of proteins through RNA, and controls the growth, development and working of cells.

(x) The two strands in DNA are not identical but are complementary. Give reason.

The two strands are complementary because their bases always pair in a fixed way: adenine pairs with thymine and guanine pairs with cytosine. Each strand therefore carries

the matching sequence of the other, so the strands are mirror images in base pairing, not identical copies.

(xi) Write the important structural and functional differences between DNA and RNA.

Structurally, DNA is a double helix with deoxyribose and thymine, while RNA is usually a single strand with ribose and uracil. Functionally, DNA stores genetic information long-term, while RNA copies that information (mRNA), carries amino acids (tRNA) and forms ribosomes (rRNA) to build proteins.

(xii) Name the different types of RNA found in the cell.

The main types of RNA in the cell are messenger RNA (mRNA), transfer RNA (tRNA) and ribosomal RNA (rRNA), with small nuclear RNA (snRNA) also present in the nucleus.

(xiii) Define the following terms: (i) Biochemistry (ii) Molecular biology (iii) Carbohydrates (iv) Proteins (v) Glycerol (vi) Protoplasm (vii) Polysaccharides (viii) Peptide bond (ix) Transcription (x) Translation.

(i) Biochemistry: study of the chemistry of living things. (ii) Molecular biology: study of life at the level of molecules, especially DNA, RNA and proteins. (iii) Carbohydrates: biomolecules of carbon, hydrogen and oxygen that give energy. (iv) Proteins: large molecules made of amino acids, used for structure, enzymes and defence. (v) Glycerol: a three-carbon alcohol that, with fatty acids, forms fats. (vi) Protoplasm: the living contents of a cell, including cytoplasm and nucleus. (vii) Polysaccharides: large carbohydrates made of many sugar units, such as starch and cellulose. (viii) Peptide bond: the bond joining two amino acids. (ix) Transcription: copying genetic information from DNA into RNA. (x) Translation: using the mRNA code at a ribosome to build a protein.

C. Extended response questions.

(i) How are amino acids linked to form proteins?

Amino acids are linked by peptide bonds: the carboxyl group of one amino acid joins the amino group of the next, with a water molecule removed. A chain of many amino acids forms a polypeptide, and one or more polypeptides fold into a specific shape to become a working protein.

(ii) What are the two kinds of subunits that make up a fat molecule, and how are they arranged in the molecule?

A fat molecule is made of one glycerol molecule and three fatty acid molecules. The three fatty acids are joined to the three carbon atoms of glycerol by ester bonds, forming a triglyceride, which stores energy in the body.

(iii) Write down the functions of proteins.

Proteins act as enzymes that speed up reactions, form structures such as muscles, skin and hair, transport substances such as haemoglobin carrying oxygen, defend the body as antibodies, and act as hormones that control body processes.

(iv) What are carbohydrates? Write down the properties of disaccharides and polysaccharides.

Carbohydrates are biomolecules of carbon, hydrogen and oxygen that supply energy. Disaccharides are made of two sugar units, are sweet and soluble in water, and are broken down by hydrolysis, for example sucrose and maltose. Polysaccharides are made of many sugar units, are usually not sweet and often insoluble, and serve as storage (starch, glycogen) or structure (cellulose).

(v) Explain the Watson and Crick model of DNA.

Watson and Crick proposed in 1953 that DNA is a double helix: two strands wound around each other, about 20 Angstrom wide, with sugar-phosphate backbones outside and nitrogenous bases inside. Adenine always pairs with thymine and guanine with cytosine, by hydrogen bonds, and the strands run in opposite directions and are complementary.

(vi) What are the three components of a nucleotide? How are nucleotides linked to form nucleic acids?

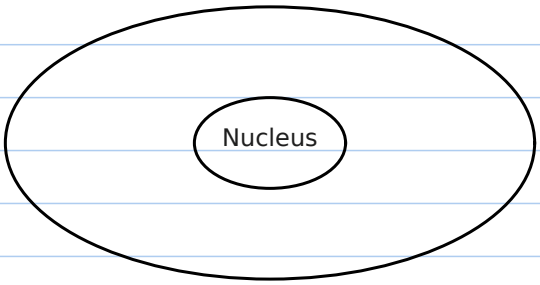
A nucleotide has three parts: a nitrogenous base, a pentose sugar (deoxyribose in DNA, ribose in RNA) and a phosphate group. Nucleotides are linked by phosphodiester bonds between the sugar of one nucleotide and the phosphate of the next, forming a long polynucleotide chain, which is a nucleic acid.

(vii) What is central dogma or gene expression? Explain.

The central dogma is the flow of genetic information: DNA -> RNA -> protein. First DNA is transcribed into messenger RNA in the nucleus, then the mRNA is translated at a ribosome, where transfer RNA brings amino acids in the order coded by the mRNA, building a protein. This expression of a gene as a protein is gene expression.

Chapter 3: Cell and Subcellular Organelles

Diagram: A generalised cell (Unit 3)



Cell membrane encloses cytoplasm and organelles

Exercise 3

A. Multiple choice questions. (Correct option marked with a green tick.)

(1) Protein factories of cells are:

(a) Ribosomes ✓

(b) Lysosomes

(c) Golgi bodies

(d) Mitochondrion

(2) The organelle discovered by Robert Brown:

(a) Ribosome

(b) Mitochondria

(c) Lysosome

(d) Nucleus ✓

(3) Mitochondria are sausage-shaped filamentous structures surrounded by:

(a) Single layered membrane

(b) Double layered membrane ✓

(c) Triple layered membrane

(d) Cellulose

(4) The percentage of lipid in cell membrane is:

(a) 10 - 15%

(b) 45 - 50% ✓

(c) 20 - 40%

(d) 60 - 80%

(5) The percentage of water in cytoplasm is:

(a) 60%

(b) 70%

(c) 80% ✓

(d) 90%

(6) Each centriole is made up of:

(a) 7 microtubules

(b) 27 microtubules ✓

(c) 17 microtubules

(d) 37 microtubules

(7) The cell membrane of vacuole is called:

(a) Protoplast

(b) Symplast

(c) Apoplast

(d) Tonoplast ✓

(8) Which one is involved in cell division and is a major component of cilia and flagella?

(a) Centrioles

(b) Golgi bodies

(c) Microtubules ✓

(d) Ribosomes

(9) The cell wall of prokaryotes contains:

(a) Lignin

(b) Peptidoglycan ✓

(c) Chitin

(d) Cellulose

(10) Nucleolus is made up of:

(a) Protein and DNA

(b) DNA and Ribosome

(c) Ribosome and protein

(d) Protein and RNA ✓

(11) Smooth muscles are found in:

(a) Heart

(b) Blood vessels ✓

(c) Liver

(d) Brain

(12) Phragmoplast is formed from:

(a) Ribosome

(b) Vesicle of ER

(c) Golgi bodies ✓

(d) Mitochondrion

(13) In which phase of meiosis does the crossing over occur?

(a) Telophase - I

(b) Metaphase - I

(c) Anaphase - I

(d) Prophase - I ✓

(14) In human beings there are 46 chromosomes (2n), what will be the number of chromosomes in their gametes?

(a) 22

(b) 23 ✓

(c) 24

(d) 25

(15) DNA replication takes place in:

(a) G1 phase

(b) S phase ✓

(c) M phase

(d) G2 phase

(16) Microtubules are formed in:

(a) G1 phase

(b) S phase

(c) M phase ✓

(d) G1 phase

(17) Partitioning of daughter chromosomes takes place in:

(a) G1 phase

(b) S phase

(c) M phase ✓

(d) G2 phase

(18) The most lengthy phase of meiosis is:

(a) Anaphase - I

(b) Metaphase - II

(c) Prophase - I ✓

(d) Prophase - II

(19) Separation of homologous chromosomes occurs during:

(a) Prophase - I

(b) Metaphase - I

(c) Telophase - I

(d) Anaphase - I ✓

B. Short response questions.

(i) Why is cell called the basic unit of life?

The cell is called the basic unit of life because all living things are made of cells, and all life processes — nutrition, respiration, growth and reproduction — take place inside cells. A single cell can carry out life on its own, and no smaller part of an organism is alive by itself.

(ii) How will you briefly describe the cytoskeleton?

The cytoskeleton is a network of protein fibres inside the cell, made of microtubules, actin filaments and intermediate filaments. It gives the cell its shape, supports organelles, helps in movement and division, and transports materials inside the cell.

(iii) What do you know about the Golgi bodies?

Golgi bodies are stacks of flat, disc-shaped sacs called cisternae. They receive proteins from the endoplasmic reticulum, modify, pack and label them, and send them to their destination inside or outside the cell. Lysosomes are also formed by Golgi bodies.

(iv) What are neurons? What do you know about the structure of a typical neuron?

Neurons are nerve cells that carry messages in the body. A typical neuron has a cell body containing the nucleus, short branches called dendrites that receive messages, and a long axon that carries messages away, often covered by a myelin sheath, ending in branches that pass the message to the next cell.

(v) What are muscle cells? Write down the types of muscles.

Muscle cells are long, contractile cells that produce movement by shortening. The three types are skeletal muscles (attached to bones, voluntary), smooth muscles (in walls of organs such as blood vessels and gut, involuntary) and cardiac muscles (in the heart, involuntary and never tired).

(vi) Briefly describe the cell specialization.

Cell specialization (differentiation) is the process by which unspecialised cells develop special structures and functions, so that different cells do different jobs, for example nerve cells conduct messages while muscle cells contract. This division of labour makes multicellular organisms work efficiently.

(vii) Write down the functions of lysosomes.

Lysosomes contain digestive enzymes that break down worn-out organelles, foreign particles and large molecules. They clean the cell, digest food taken in by the cell, and destroy harmful bacteria, acting as the cell's digestive and cleaning system.

(viii) What are the differences between animal and plant cells?

Plant cells have a cell wall, large central vacuole and chloroplasts, and usually no centrioles, while animal cells have no cell wall or chloroplasts, have small vacuoles or none, and have centrioles. Both have a cell membrane, cytoplasm, nucleus and mitochondria.

(ix) Differentiate between smooth and rough endoplasmic reticulum.

Rough endoplasmic reticulum has ribosomes on its surface, looks rough, and makes and transports proteins. Smooth endoplasmic reticulum has no ribosomes, looks smooth, and makes lipids, detoxifies substances and stores calcium.

(x) Distinguish cytokinesis from karyokinesis.

Karyokinesis is the division of the nucleus, in which chromosomes separate into two sets.

Cytokinesis is the division of the cytoplasm that follows, splitting one cell into two daughter cells. Karyokinesis divides the genetic material; cytokinesis divides the cell itself.

(xi) Describe the events taking place during G1 phase of interphase.

During the G1 (Gap 1) phase the cell grows, makes new organelles and proteins, and carries out its normal functions. It also checks that conditions are suitable and prepares the materials needed before DNA replication begins in the S phase.

(xii) How does cytokinesis in plant cells differ from that in animal cells?

In animal cells cytokinesis happens by a cleavage furrow: the cell membrane pinches inward until the cell splits into two. In plant cells, which have a rigid wall, a cell plate (phragmoplast) forms across the middle from Golgi vesicles and grows into a new cell wall dividing the cell.

(xiii) What is quiescent stage (G0)?

The G0 stage is a resting stage in which a cell leaves the cell cycle and stops dividing, either temporarily or permanently. Many specialised cells, such as nerve cells, remain in G0, but some can re-enter the cycle when needed.

(xiv) What is the significance of meiosis?

Meiosis halves the chromosome number so that gametes have half the chromosomes (n) and fertilisation restores the full number (2n). It also creates variation through crossing over and independent assortment, which is important for evolution and adaptation.

(xv) Distinguish between metaphase of mitosis and metaphase-I of meiosis.

In metaphase of mitosis, single chromosomes line up singly on the equator and sister chromatids will separate. In metaphase-I of meiosis, homologous chromosome pairs line up together on the equator, and whole homologous chromosomes (not chromatids) will separate in anaphase-I.

(xvi) Define the following terms: (i) Cell cycle (ii) Granum (iii) Plasmodesmata (iv) Synapsis (v) G1 phase (Gap 1) (vi) Chiasmata (vii) Centromere (viii) G2 phase (Gap 2) (ix) Meiosis (x) Mitosis (xi) Crossing over (xii) S-phase (xiii) Kinetochore (xiv) Spindle fibres (xv) Phragmoplast.

(i) Cell cycle: the ordered series of events from one cell division to the next, including interphase and mitotic phase. (ii) Granum: a stack of thylakoids in a chloroplast. (iii) Plasmodesmata: tiny channels through plant cell walls connecting neighbouring cells. (iv) Synapsis: pairing of homologous chromosomes in prophase-I of meiosis. (v) G1 phase: the first gap phase of interphase, for growth and preparation. (vi) Chiasmata: points where non-sister chromatids cross and exchange segments. (vii) Centromere: the region joining sister chromatids, where spindle fibres attach. (viii) G2 phase: the second gap phase, for final preparation before division. (ix) Meiosis: division producing four haploid cells from one diploid cell. (x) Mitosis: division producing two genetically identical diploid cells. (xi) Crossing over: exchange of segments between non-sister chromatids of homologous

chromosomes. (xii) S-phase: the synthesis phase in which DNA is replicated. (xiii) Kinetochore: the protein structure on a centromere where spindle fibres attach. (xiv) Spindle fibres: microtubule fibres that pull chromosomes apart during division. (xv) Phragmoplast: the structure forming the cell plate during plant cytokinesis.

C. Extended response questions.

(i) Write a detailed note on plastids.

Plastids are membrane-bound organelles found only in plant cells, usually containing pigments that give plant cells their colours. Chloroplasts contain chlorophyll and carry out photosynthesis. Chromoplasts contain coloured pigments and give colour to flowers and fruits. Leucoplasts are colourless and store food such as starch, proteins and oils. All plastids have their own DNA and double membrane, and they develop from proplastids.

(ii) Who discovered mitochondria? Explain its structure. Why are they called the powerhouse of the cell?

Mitochondria were first observed by Albert von Kolliker and named by Carl Benda. A mitochondrion is a sausage-shaped, double-membrane organelle: the outer membrane is smooth, the inner membrane folds into cristae, and the inner space (matrix) contains enzymes and its own DNA. Mitochondria release energy from food by aerobic respiration, producing ATP, the cell's energy currency, so they are called the powerhouse of the cell.

(iii) Write a comprehensive note on nucleus.

The nucleus, discovered by Robert Brown, is the control centre of the cell, often called its brain. It is bounded by a double nuclear membrane with pores, contains nucleoplasm, a nucleolus that makes ribosomal RNA, and chromatin (DNA and protein) that condenses into chromosomes during division. The nucleus stores genetic information and controls all cell activities, including growth and division.

(iv) Explain epidermal and mesophyll cells.

Epidermal cells form the outer protective layer (epidermis) of leaves and other plant parts; they are flat, closely packed, usually without chloroplasts, and covered by a waxy cuticle that reduces water loss, with stomata for gas exchange. Mesophyll cells are the inner leaf cells packed with chloroplasts: palisade mesophyll cells are tall and arranged in a layer for maximum photosynthesis, while spongy mesophyll cells are loosely arranged with air spaces for gas exchange.

(v) Write a detailed note on the division of labour within a cell and across cells.

Within a cell, labour is divided among organelles: the nucleus controls, mitochondria supply energy, ribosomes make proteins, Golgi bodies pack them, lysosomes clean, and the membrane controls entry and exit. Across cells, different specialised cells do different jobs — nerve cells conduct messages, muscle cells contract, red blood cells carry oxygen — and tissues, organs and systems divide labour further, so the whole organism works efficiently.

(vi) Write a note on the structural advantages of plant and animal cells.

Plant cells gain advantage from a rigid cell wall that gives support and protection, a large vacuole that stores water and maintains turgor, and chloroplasts that make their own food. Animal cells gain advantage from having no rigid wall, so they can change shape and move, from centrioles that organise division, and from small vacuoles and flexible membranes suited to movement and varied functions.

(vii) Give a comparison between mitosis and meiosis.

Mitosis occurs in body (somatic) cells, divides once, and produces two genetically identical diploid (2n) cells, for growth and repair. Meiosis occurs in reproductive cells, divides twice, and produces four genetically different haploid (n) cells, for sexual reproduction. Meiosis includes crossing over and independent assortment, creating variation; mitosis does not.

(viii) Describe the stages of meiosis-I with labelled diagram.

Meiosis-I stages: Prophase-I — chromosomes condense, homologous chromosomes pair (synapsis), crossing over occurs at chiasmata. Metaphase-I — homologous pairs line up on the equator. Anaphase-I — homologous chromosomes separate to opposite poles. Telophase-I — two haploid cells form, each with half the chromosomes. Labelled diagram: draw two cells showing paired chromosomes at the equator in metaphase-I, then separating in anaphase-I, with spindle fibres, poles and cell membrane labelled.

(ix) Explain the stages of mitosis with labelled diagram.

Mitosis stages: Prophase — chromosomes condense and the spindle forms. Metaphase — chromosomes line up singly on the equator. Anaphase — sister chromatids separate to opposite poles. Telophase — nuclear membranes reform around two identical sets of chromosomes, followed by cytokinesis. Labelled diagram: draw a cell at each stage showing chromosomes, spindle fibres, centrioles and equator, with each part labelled.

Chapter 4: Tissues, Organs and Systems

Exercise 4

A. Multiple choice questions. (Correct option marked with a green tick.)

(1) A group of cells having similar structure and function is called:

(a) Organ

(b) Tissue ✓

(c) Organ system

(d) Organism

(2) The living sub-cellular bodies in a cell are called:

(a) Inclusions

(b) Tissue

(c) Organelles ✓

(d) Organ

(3) Which statement is not correct during playing (exercise)?

(a) More heat is produced

(b) Increase in blood flow

(c) More O_2 is released ✓

(d) More CO_2 is released

(4) The spongy mesophyll cells are:

(a) Round in shape

(b) Triangular in shape

(c) Column shaped

(d) Irregular in shape ✓

(5) Liver breaks down the:

(a) Dead WBCs

(b) Dead RBCs ✓

(c) Platelets

(d) Plasma

(6) Zygote after repeated divisions forms:

(a) Seed

(b) Embryo ✓

(c) Root

(d) Leaf

(7) Which one of the following contains chlorophyll?

(a) Epidermal cells

(b) Flower

(c) Guard cells ✓

(d) Xylem tissues

(8) The disorder of nervous system is:

(a) Ulcer

(b) pneumonia

(c) Tumor ✓

(d) goiter

(9) One of the following is not nitrogenous waste?

(a) Ammonia

(b) Uric acid

(c) Bilirubin ✓

(d) Urea

(10) In the skin, the thin layer of fat cells is called:

(a) Cutin

(b) Dermis

(c) Adipose tissue ✓

(d) Epidermis

B. Short response questions.

(i) What are the advantages of homeostasis?

Homeostasis keeps the body's internal conditions, such as temperature, water and salts, steady despite outside changes. This lets enzymes work at their best, keeps cells alive and healthy, and allows the body to function normally in hot, cold or dry conditions.

(ii) Justify that lungs are homeostatic organs.

Lungs are homeostatic organs because they control the levels of oxygen and carbon dioxide in the blood: they take in oxygen needed for respiration and remove carbon dioxide, a waste product. By adjusting breathing rate, they keep blood gases and pH within narrow, healthy limits.

(iii) What is the role of skin in osmoregulation and thermoregulation?

In osmoregulation the skin removes water and salts as sweat through sweat glands, helping balance body water. In thermoregulation the skin loses heat by sweating and by widening blood vessels in hot weather, and conserves heat by narrowing vessels and raising goose bumps in cold weather.

(iv) Define the endocrine glands and write their role in human body.

Endocrine glands are ductless glands that release hormones directly into the blood. Hormones act as chemical messengers that control growth, metabolism, reproduction, blood sugar and mood, coordinating body processes over longer periods than nerves do.

(v) Define the following terms: (i) Organ (ii) Epidermis (iii) Midrib (iv) Respiratory System (v) Excretory system (vi) Homeostasis (vii) Goose bumps (viii)

Thermoregulation (ix) Embryo (x) Fruit.

(i) Organ: a structure made of different tissues working together for one function, such as the heart. (ii) Epidermis: the outermost protective layer of skin or leaf. (iii) Midrib: the central vein of a leaf that supports it and carries water and food. (iv) Respiratory System: the organs, including nose, trachea and lungs, that take in oxygen and remove carbon dioxide. (v) Excretory system: the organs, including kidneys, skin and lungs, that remove wastes from the body. (vi) Homeostasis: keeping internal conditions steady. (vii) Goose bumps: raised skin caused by tiny muscles pulling hairs upright to trap air and conserve heat. (viii) Thermoregulation: controlling body temperature. (ix) Embryo: the early developing organism formed from a zygote by repeated divisions. (x) Fruit: the ripe ovary with seeds, formed after fertilisation.

C. Extended response questions.

(i) Explain the levels of organization in living organisms.

The levels of organization are: cell (basic unit), tissue (group of similar cells), organ (different tissues working together), organ system (organs working together), and organism (all systems together). For example, muscle cells form muscle tissue, which with other tissues forms the heart, which with blood vessels forms the circulatory system of a human being.

(ii) Write a detailed note on composition of stomach. (Note: the book chapter is on tissues, organs and systems; the stomach is treated here as an organ example.)

The stomach is an organ of the digestive system, made of four tissue layers: the outer serosa, a muscle layer of smooth muscle that churns food, a submucosa with blood vessels and nerves, and an inner mucosa that secretes gastric juice (acid and enzymes) and mucus. Together these tissues let the stomach store, mix and partly digest food.

(iii) Explain the composition of a leaf on tissue level.

A leaf is made of upper and lower epidermis (protective layers, with stomata and guard cells in the lower layer), mesophyll tissue between them (palisade mesophyll for photosynthesis and spongy mesophyll with air spaces for gas exchange), and vascular tissue (xylem bringing water and phloem carrying food) in veins, with the midrib supporting the leaf.

(iv) Discuss the roles of various plant organs in plant physiology.

Roots absorb water and minerals and anchor the plant. Stems support the plant and transport water (xylem) and food (phloem) between roots and leaves. Leaves make food by photosynthesis and exchange gases through stomata. Flowers are the reproductive organs, producing seeds, and fruits protect and spread seeds.

(v) Write a note on human nervous system.

The human nervous system has the central nervous system (brain and spinal cord) and the

peripheral nervous system (nerves). It receives information through sense organs, processes it in the brain, and sends orders to muscles and glands through motor nerves, coordinating rapid responses. Disorders of the nervous system include tumours and injuries that disturb this coordination.

Chapter 5: Metabolism

Exercise 5

A. Multiple choice questions. (Correct option marked with a green tick.)

(1) The induced fit model for the mechanism of enzyme action was proposed by:

(a) Emil Fischer

(b) Singer and Nicolson

(c) Daniel Koshland ✓

(d) Schleiden and Schwann

(2) The function of ATP is:

(a) Storage of Energy ✓

(b) Generation of Energy

(c) Consumption of Energy

(d) Transformation of Energy

(3) Anaerobic respiration takes place in:

(a) Cytoplasm ✓

(b) Endoplasmic Reticulum

(c) Nucleus

(d) Mitochondria

(4) The products of alcoholic fermentation are:

(a) Lactic acid and ethanol

(b) Lactic acid and CO_2

(c) Ethanol and CO_2 ✓

(d) Ethanol and CO_2

(5) Raw materials for photosynthesis are:

(a) Water and glucose

(b) CO_2 and glucose

(c) Glucose and chlorophyll

(d) Water and CO_2 ✓

(6) During aerobic respiration the net gain of ATP is:

(a) 34

(b) 36 ✓

(c) 38

(d) 40

(7) The number of ATP molecules obtained from one molecule of glucose in anaerobic respiration are:

(a) 5

(b) 4

(c) 3

(d) 2 ✓

(8) The primary purpose of catabolism in cellular metabolism is to:

(a) Synthesize new biomolecules

(b) Generate energy for cellular processes ✓

(c) Store energy for future use

(d) Repair damaged tissues

(9) The lock and key model of enzyme action states that:

(a) Enzymes change their shape to fit substrates

(b) Enzymes and substrates have specific shapes ✓

(c) Enzymes are non-specific and bind to any substrate

(d) Enzymes are denatured by substrate

(10) The induced fit model of enzyme action states that:

(a) Enzymes remain rigid and unchanged

(b) Enzymes change shape to fit substrate ✓

(c) Substrate changes shape to fit enzyme

(d) Enzymes are non-specific and bind to any substrate

(11) When an inhibitor binds to the active site of an enzyme this inhibition is called:

(a) Non-competitive inhibition

(b) Competitive inhibition ✓

(c) Allosteric inhibition

(d) Irreversible inhibition

(12) The products of light reactions of photosynthesis are:

(a) Glucose and oxygen

(b) ATP and NADPH ✓

(c) Water and Carbon dioxide

(d) Chlorophyll and light

(13) The difference between light reactions and dark reactions in terms of location is:

(a) Light reactions occur in stroma, dark reactions occur in thylakoid membrane

(b) Light reactions occur in thylakoid membrane, dark reactions occur in stroma ✓

(c) Both reactions occur in stroma

(d) Both reactions occur in thylakoid membrane

(14) The three stages of aerobic respiration are:

(a) Glycolysis, fermentation and electron transport chain

(b) Glycolysis, pyruvate oxidation and Krebs cycle

(c) Glycolysis, Krebs cycle and electron transport chain ✓

(d) Glycolysis, photosynthesis and Krebs cycle

B. Short response questions.

(i) What is the difference between catabolism and anabolism?

Catabolism breaks down large molecules into smaller ones and releases energy, for example respiration breaking glucose into carbon dioxide and water. Anabolism builds large molecules from smaller ones and uses energy, for example photosynthesis building glucose from water and carbon dioxide.

(ii) What is an active site and how is it important?

The active site is the specific pocket on an enzyme where the substrate binds. Its shape fits only a particular substrate, like a lock fits a key, so the enzyme acts only on that substrate and speeds up only its reaction.

(iii) What is the importance of photosynthesis?

Photosynthesis makes food for almost all life, directly or indirectly, produces the oxygen we breathe, and reduces carbon dioxide in the air. Without it, food chains would collapse and the atmosphere would lose oxygen.

(iv) Write the mechanism of enzyme action.

First the substrate binds to the enzyme's active site, forming an enzyme-substrate (E-S) complex. The enzyme strains the substrate's bonds (and, in the induced fit model, changes shape slightly to fit it better), the reaction happens, products are released, and the enzyme is unchanged and ready to act again.

(v) Write a brief note on fermentation.

Fermentation is anaerobic respiration, in which glucose is broken down without oxygen, giving only 2 ATP per glucose. In yeast it produces ethanol and carbon dioxide (alcoholic fermentation, used in baking and brewing); in muscle cells it produces lactic acid, which causes tiredness.

(vi) Differentiate between aerobic and anaerobic respiration.

Aerobic respiration uses oxygen, takes place in cytoplasm and mitochondria, fully breaks glucose into carbon dioxide and water, and gives 36 ATP. Anaerobic respiration does not use oxygen, takes place in cytoplasm only, partly breaks glucose into ethanol or lactic acid, and gives only 2 ATP.

(vii) Why is ATP the major energy currency of all living cells?

ATP is the major energy currency because energy released from food is stored in its high-energy phosphate bonds and released in small, usable amounts whenever a cell needs it, for movement, transport, synthesis and growth. ATP is made in respiration and photosynthesis and used by every cell.

(viii) Interpret the role of chlorophyll in relation with light in photosynthesis.

Chlorophyll is the green pigment in chloroplasts that absorbs sunlight, mainly blue and red

light, and reflects green light. The absorbed light energy drives the light reactions: splitting water (photolysis), releasing oxygen, and making ATP and NADPH for the dark reactions.

(ix) How do plants get water and carbon dioxide for photosynthesis?

Plants get water from the soil through root hairs, carried up by xylem to the leaves. They get carbon dioxide from the air through stomata, tiny pores in the leaf epidermis controlled by guard cells.

(x) Write down the process of photosynthesis and respiration in the form of equations.

Photosynthesis: 6CO2 + 6H2O + light energy -> C6H12O6 + 6O2 (in chloroplasts). Aerobic respiration: C6H12O6 + 6O2 -> 6CO2 + 6H2O + energy (36 ATP). The two processes are opposite: photosynthesis stores energy, respiration releases it.

(xi) Define the following terms: (i) Metabolism (ii) Substrate (iii) Enzyme (iv) E-S complex (v) Active site (vi) Optimum pH of enzyme (vii) Inhibitors (viii) ATP (ix) Photosynthesis (x) Fermentation (xi) Aerobic respiration (xii) Glycolysis (xiii) Denaturation of enzyme (xiv) Photolysis of water (xv) Electron transport chain.

(i) Metabolism: all chemical reactions in a cell. (ii) Substrate: the substance an enzyme acts on. (iii) Enzyme: a biological catalyst, usually a protein, that speeds up a reaction without being used up. (iv) E-S complex: the temporary enzyme-substrate combination. (v) Active site: the part of an enzyme where the substrate binds. (vi) Optimum pH: the pH at which an enzyme works fastest. (vii) Inhibitors: substances that slow or stop enzyme action. (viii) ATP: adenosine triphosphate, the cell's energy currency. (ix) Photosynthesis: making glucose from water and carbon dioxide using light energy. (x) Fermentation: anaerobic breakdown of glucose into ethanol or lactic acid. (xi) Aerobic respiration: breakdown of glucose using oxygen, giving carbon dioxide, water and much energy. (xii) Glycolysis: the first stage of respiration, splitting glucose into pyruvate in the cytoplasm. (xiii) Denaturation: loss of an enzyme's shape (and function) by heat or extreme pH. (xiv) Photolysis: splitting of water by light in photosynthesis, releasing oxygen. (xv) Electron transport chain: a series of carriers in mitochondria (and chloroplasts) that transfer electrons and make ATP.

C. Extended response questions.

(i) Describe lock and key model and induced fit model for mechanism of enzyme action.

Lock and key model (Emil Fischer): the enzyme's active site has a fixed shape that exactly fits its substrate, like a key fits a lock, so only that substrate can bind and react. Induced fit model (Daniel Koshland): the active site is flexible and changes shape slightly when the substrate binds, fitting it more closely and straining its bonds so the reaction happens faster. Both models explain enzyme specificity; induced fit also explains how enzymes actually speed up reactions.

(ii) List down the characteristics of enzymes.

Enzymes are biological catalysts made of protein; they speed up reactions without being used up; they are specific, each acting on one substrate; they work best at an optimum temperature and pH; they are needed in small amounts; and they can be inhibited or denatured by heat, extreme pH or inhibitors.

(iii) Explain the factors which affect the enzymatic action.

Temperature: rate rises with temperature up to an optimum, then falls as the enzyme denatures. pH: each enzyme has an optimum pH; extremes denature it. Substrate concentration: rate rises until all active sites are full. Enzyme concentration: more enzyme gives a faster rate. Inhibitors slow or block action, and cofactors or coenzymes may be needed for activity.

(iv) Write down the events of light and dark reactions.

Light reactions occur in thylakoid membranes: chlorophyll absorbs light, water is split (photolysis) releasing oxygen, and ATP and NADPH are made. Dark reactions (Calvin cycle) occur in the stroma: carbon dioxide is fixed using ATP and NADPH, and glucose is built up in a cycle of reactions that does not need light directly.

(v) Explain the mechanism of aerobic respiration.

Aerobic respiration has three stages. Glycolysis in the cytoplasm splits glucose into pyruvate, giving a small ATP gain. Pyruvate enters mitochondria, is oxidised, and runs through the Krebs cycle, releasing carbon dioxide and hydrogen carriers. The electron transport chain then transfers electrons, uses oxygen as the final acceptor to form water, and makes most of the ATP — 36 ATP per glucose in total.

(vi) Compare photosynthesis and respiration.

Photosynthesis occurs in chloroplasts of green plants, uses light, water and carbon dioxide, makes glucose and oxygen, and stores energy. Respiration occurs in mitochondria of all living cells, uses glucose and oxygen, makes carbon dioxide and water, and releases energy as ATP. They are opposite and complementary processes that together cycle carbon, oxygen and energy.

Chapter 6: Biodiversity and Classification

Exercise 6

A. Multiple choice questions. (Correct option marked with a green tick.)

(1) Who is called the father of science?

(a) Theophrastus

(b) Carolus Linnaeus

(c) Robert Whittaker

(d) Aristotle ✓

(2) The classification of living organisms by Aristotle based on:

(a) Respiration

(b) Excretion

(c) Habitat ✓

(d) Mode of nutrition

(3) Meaning of biodiversity is:

(a) Diversity of plants

(b) Diversity of animals

(c) Diversity of viruses

(d) Diversity of life ✓

(4) Five kingdoms system was introduced by:

(a) Robert Brown

(b) Robert Whittaker ✓

(c) Aristotle

(d) Carolus Linnaeus

(5) Rana tigrina is the scientific name of:

(a) Cat

(b) Frog ✓

(c) Pea

(d) Human beings

(6) Zea mays is the scientific name of:

(a) Rice

(b) Wheat

(c) Frog

(d) Corn ✓

(7) The branch of biology that deals with the classification and naming of organisms is:

(a) Morphology

(b) Anatomy

(c) Physiology

(d) Taxonomy ✓

(8) Three domain system of organisms is introduced by:

(a) Aristotle

(b) Robert Whittaker

(c) Carolus Linnaeus

(d) Carl Woese ✓

(9) The term Tracheophytes is used for:

(a) Non-vascular plants

(b) Vascular plants ✓

(c) Organisms without proper nucleus

(d) Those organisms who are not clearly fit into plants and animals

B. Short response questions.

(i) Do you think that classification is necessary for the proper study of organisms? Briefly describe your answer with some facts.

Yes, classification is necessary because millions of species exist and cannot be studied one by one without order. Grouping organisms by similarities lets us identify them, study their relationships, predict their characters, and communicate about them worldwide with one scientific name.

(ii) Write down the biological names of pea, potato and corn.

Pea: Pisum sativum. Potato: Solanum tuberosum. Corn (maize): Zea mays.

(iii) Why is it difficult to classify the viruses with other organisms?

Viruses are difficult to classify because they are on the border of living and non-living: they have no cells, cannot carry out metabolism on their own, and only reproduce inside a host cell. They therefore do not fit into any kingdom of cellular organisms.

(iv) Define the following terms: (i) Prokaryotes (ii) Hierarchy (iii) Species (iv) Taxonomy (v) Binomial nomenclature (vi) Biodiversity.

(i) Prokaryotes: organisms whose cells lack a true nucleus and membrane-bound organelles, such as bacteria. (ii) Hierarchy: the ranked levels of classification, from domain down to species. (iii) Species: a group of similar organisms that can interbreed and produce fertile offspring. (iv) Taxonomy: the branch of biology dealing with classification and naming of organisms. (v) Binomial nomenclature: the method of giving each organism a two-word scientific name (genus + species). (vi) Biodiversity: the variety of life, including genes, species and ecosystems.

C. Extended response questions.

(i) What is five kingdom classification system for living organisms? Who introduced this system? Who modified the five kingdom system?

The five kingdom system divides living things into Monera (prokaryotes), Protista, Fungi, Plantae and Animalia, based on cell type, body organisation and mode of nutrition. It was introduced by Robert Whittaker in 1969 and later modified when the three-domain system (Archaea, Bacteria, Eukarya) was proposed by Carl Woese, separating archaea from bacteria.

(ii) Write a comprehensive note on three distinct domains of organisms.

The three domains are Archaea (ancient prokaryotes living in extreme environments, with distinct cell chemistry), Bacteria or Eubacteria (common prokaryotes such as disease and soil bacteria), and Eukarya (all organisms with a true nucleus, including protists, fungi, plants and animals). Domains are the highest rank of classification, above kingdoms.

(iii) Explain in detail the taxonomic ranks of classification.

The taxonomic ranks, from largest to smallest, are: Domain, Kingdom, Phylum (Division in plants), Class, Order, Family, Genus and Species. Each lower rank includes fewer, more similar organisms; for example humans belong to domain Eukarya, kingdom Animalia, phylum Chordata, class Mammalia, order Primates, family Hominidae, genus Homo and species sapiens.

(iv) Who introduced binomial nomenclature? What are the rules for binomial nomenclature? What is the importance of binomial nomenclature?

Binomial nomenclature was introduced by Carolus Linnaeus. Rules: every organism gets two names, genus (capitalised) and species (small letters), written in italics (or underlined when handwritten), and the genus name may be abbreviated after first use. Its importance is that it gives every organism one universal scientific name, avoiding confusion from different local names in different languages.

Chapter 7: Form and Function in Plants

Exercise 7

A. Multiple choice questions. (Correct option marked with a green tick.)

(1) The essential component of an amino acid is:

(a) Magnesium

(b) Sulphur

(c) Nitrogen ✓

(d) Sodium

(2) In plants, food is stored in the form of:

(a) Glucose

(b) Starch ✓

(c) Sucrose

(d) Proteins

(3) The estimated percentage of transpiration through stomata is:

(a) 1-2%

(b) 5%

(c) 50-60%

(d) 90% ✓

(4) The translocation of food in phloem is in the form of:

(a) Maltose

(b) Glucose

(c) Sucrose ✓

(d) Fructose

(5) Transpiration is not affected by:

(a) Wind

(b) Temperature

(c) Humidity

(d) Gravity ✓

(6) The direction of food transport in plants is:

(a) Upward

(b) Downward

(c) Horizontal

(d) From source towards sink ✓

(7) One of the following is not a type of asexual reproduction.

(a) Binary fission

(b) Budding

(c) Gametogenesis ✓

(d) Grafting

(8) The plants which store water in their bodies are called:

(a) Succulents ✓

(b) Mesophytes

(c) Aquatic

(d) Halophytes

(9) The vegetative structures of plants which store nutrients in fleshy stems surrounded by scale-like leaves are called:

(a) Bulbs ✓

(b) Suckers

(c) Corms

(d) Rhizomes

(10) The role of Mg in plant growth is:

(a) It is necessary for seed and flower development

(b) It is an important component of chlorophyll ✓

(c) It is required for cell division and growth

(d) It improves root growth and seed production

(11) The primary function of root hairs is:

(a) To anchor the plant

(b) To absorb water and minerals ✓

(c) To transport food

(d) To produce food for plant

(12) The mechanism of translocation in phloem tissue is called:

(a) Active transport

(b) Passive transport

(c) Pressure flow hypothesis ✓

(d) Diffusion

(13) The adaptation made by plants for osmoregulation is:

(a) Closing of stomata during drought ✓

(b) Increase transpiration rate

(c) Reduce root growth

(d) Increase photosynthesis rate

(14) The common example of artificial vegetative propagation?

(a) Bulb

(b) Grafting ✓

(c) Rhizome

(d) Tuber

(15) One of the following plants is commonly propagated by stem cutting.

(a) Rose

(b) Sugar cane

(c) Money plant

(d) All of the above ✓

(16) Cloning is a process of:

(a) Creating new species

(b) Creating an exact copy of the organism ✓

(c) Mixing genes from different organisms

(d) Evolving an organism over time

B. Short response questions.

(i) Briefly explain macronutrients.

Macronutrients are mineral nutrients that plants need in large amounts: nitrogen, phosphorus, potassium, calcium, magnesium and sulphur. They build proteins, nucleic acids and chlorophyll, and support growth, so their shortage causes poor growth and yellowing.

(ii) What are the functions of phosphorus and potassium in plants?

Phosphorus is needed for ATP, nucleic acids and cell membranes, and for root growth and flowering. Potassium controls opening and closing of stomata, activates enzymes, and helps in water balance and disease resistance.

(iii) Write down the two functions each of boron and iron in plants.

Boron: needed for cell wall formation and for flowering and seed setting. Iron: needed for chlorophyll formation (though it is not part of chlorophyll) and for electron transport in respiration and photosynthesis.

(iv) What is the role of Mg in chlorophyll formation?

Magnesium is the central atom of the chlorophyll molecule: each chlorophyll molecule holds one magnesium atom at its centre. Without magnesium, plants cannot make chlorophyll, so leaves turn yellow and photosynthesis stops.

(v) How are metabolic wastes (except water and gases) removed from plants?

Plants store solid wastes in old leaves that fall off, in bark that peels away, and in vacuoles as crystals or gums and resins. Some wastes are also removed in fruits and seeds that drop from the plant.

(vi) Briefly describe the internal structure of root.

A root in cross-section has an outer epidermis with root hairs, a cortex of parenchyma cells for storage, an endodermis controlling entry, a pericycle that gives rise to branch roots, and a central vascular cylinder with xylem and phloem for transport.

(vii) Write a short note on transpiration.

Transpiration is the loss of water vapour from a plant, mainly through stomata in leaves. It is driven by evaporation and pulls water up from the roots, cools the leaf, and helps

transport minerals, but excessive transpiration can wilt the plant.

(viii) What are the directions of food transport in plants?

Food (sucrose) is transported in phloem from source to sink: from leaves (source) to roots, fruits, seeds and growing parts (sinks). The direction can change with the season and the plant's needs, for example from storage organs to growing shoots in spring.

(ix) What is the significance of transpiration?

Transpiration pulls water and dissolved minerals up from roots to leaves (transpiration pull), cools the plant, maintains turgor and the flow of water, and helps distribute minerals. It is essential for the ascent of sap, though it also risks water loss.

(x) Differentiate natural and artificial propagation.

Natural propagation happens without human help, for example by bulbs, rhizomes, tubers, runners or suckers. Artificial propagation is done by humans, for example by cuttings, layering, grafting and tissue culture, to produce desired plants quickly and in large numbers.

(xi) Write a short note on cloning.

Cloning is producing an exact genetic copy of an organism, for example a plant grown from a cutting or by tissue culture. The clone has the same genes as the parent, so it keeps the parent's characters, which is useful for multiplying good varieties.

(xii) Write a note on root hairs.

Root hairs are tiny, thin extensions of root epidermal cells that greatly increase the absorbing surface of the root. They absorb water and mineral salts from the soil by osmosis and diffusion, and are short-lived, being replaced as the root grows.

(xiii) What are the effects of opening and closing of stomata on transpiration?

When stomata open, water vapour escapes freely and transpiration is high, while gases for photosynthesis also enter. When stomata close, transpiration falls sharply, conserving water, but photosynthesis also slows because carbon dioxide cannot enter.

(xiv) What is homeostasis? Write down its importance.

Homeostasis is keeping internal conditions steady, such as water content, salts and temperature, despite outside changes. In plants it is achieved by stomatal control, osmoregulation and storage, and it keeps cells working and enzymes active.

(xv) What are halophytes? Write down their adaptations.

Halophytes are plants that grow in salty soils or water, such as mangroves. Their adaptations include salt glands that excrete salt, fleshy leaves that dilute salt, deep roots, and seeds that germinate only when washed by fresh water.

(xvi) Define the following terms: (i) Herbivores (ii) Macronutrients (iii) Micronutrients (iv) Nutrition (v) Cuticle (vi) Transpirational pull (vii) Osmotic pressure (viii) Transpiration (ix) Cortex (x) Guard cells (xi) Sink (xii) Pericycle (xiii) Source (xiv) Cohesion (xv) Reproduction (xvi) Pollination (xvii) Embryo

(xviii) Budding (xix) Fertilization (xx) Artificial propagation.

(i) Herbivores: animals that eat only plants. (ii) Macronutrients: nutrients needed in large amounts. (iii) Micronutrients: nutrients needed in small amounts. (iv) Nutrition: taking in and using food for growth and energy. (v) Cuticle: the waxy outer layer on leaves that reduces water loss. (vi) Transpirational pull: the upward pull on water caused by transpiration from leaves. (vii) Osmotic pressure: the pressure needed to stop osmosis into a solution. (viii) Transpiration: loss of water vapour from a plant, mainly through stomata. (ix) Cortex: the tissue between epidermis and vascular cylinder in roots and stems. (x) Guard cells: bean-shaped cells that open and close a stoma. (xi) Sink: a part receiving food, such as a root or fruit. (xii) Pericycle: the layer just inside the endodermis that forms branch roots. (xiii) Source: a part producing or releasing food, such as a leaf. (xiv) Cohesion: attraction between water molecules, holding the water column together. (xv) Reproduction: producing new individuals of the same kind. (xvi) Pollination: transfer of pollen from anther to stigma. (xvii) Embryo: the young plant inside a seed. (xviii) Budding: asexual reproduction in which a small outgrowth grows into a new individual. (xix) Fertilization: fusion of male and female gametes to form a zygote. (xx) Artificial propagation: human methods of vegetative reproduction such as cuttings and grafting.

C. Extended response questions.

(i) Explain the cohesion-tension theory for ascent of sap.

The cohesion-tension theory says water lost by transpiration from leaves creates a pull (tension) that draws the water column upward in xylem. Cohesion (attraction between water molecules) keeps the column unbroken, and adhesion to xylem walls supports it, so water rises from roots to the highest leaves without pumping.

(ii) How is food translocated in plants? Explain with the help of pressure flow mechanism.

Food (sucrose) is made in leaves (source) and loaded into phloem sieve tubes, drawing water in by osmosis and raising pressure. This pressure pushes the sap towards sinks (roots, fruits), where sucrose is unloaded and water leaves, lowering pressure. The pressure difference between source and sink drives the flow, as described by the pressure flow hypothesis.

(iii) Illustrate the mechanism of opening and closing of stomata.

Guard cells control the stoma: when they take in potassium ions and water, they become turgid, their thin outer walls stretch more than their thick inner walls, and the stoma opens. When they lose water, they become flaccid and the stoma closes. Light, carbon dioxide level and water availability control this movement.

(iv) How does uptake of water and minerals occur through roots?

Root hairs absorb water by osmosis, since root cell sap is more concentrated than soil water, and minerals by diffusion and active transport using energy. Water and minerals

then move through the cortex, pass the endodermis, and enter xylem vessels for transport upward.

(v) What are factors affecting the rate of transpiration?

Transpiration increases with higher temperature, stronger wind, lower humidity and brighter light (which opens stomata). It decreases in humid, still, cool or dark conditions, and with a thick cuticle, fewer stomata or closed stomata. Leaf area and water availability also affect it.

(vi) What are different methods of asexual reproduction in plants? Explain briefly.

Asexual reproduction in plants includes vegetative propagation by roots, stems and leaves (runners, rhizomes, bulbs, tubers, suckers), budding, fragmentation, and spore formation. In each, a new plant grows from a part of the parent without seeds or gametes, so it is genetically identical to the parent.

(vii) Explain different types of artificial vegetative propagation in plants.

Artificial vegetative propagation includes cutting (a stem, root or leaf piece is planted, e.g. rose, sugarcane), layering (a branch is bent to the soil until it roots), grafting (a scion is joined to a rootstock of another plant), and tissue culture (small pieces grown in sterile nutrient medium into many plantlets).

(viii) Define flower and explain its different parts.

A flower is the reproductive part of a plant. Its parts are: sepals (calyx, protecting the bud), petals (corolla, attracting pollinators), stamens (male parts: anther producing pollen, on a filament), and carpels (female parts: stigma receiving pollen, style, and ovary containing ovules that become seeds).

(ix) What is vegetative propagation? Describe its types.

Vegetative propagation is asexual reproduction in which a new plant grows from a vegetative part (root, stem or leaf) of the parent. Its types are natural (by runners, rhizomes, bulbs, tubers, suckers and leaves) and artificial (by cuttings, layering, grafting and tissue culture). The new plants are genetically identical to the parent.

Chapter 8: Evolution and Natural Selection

Exercise 8

A. Multiple choice questions. (Correct option marked with a green tick.)

(1) Aristotle classified the organisms on the basis of:

(a) Respiration

(b) Excretion

(c) Reproduction

(d) Mode of nutrition ✓

(2) The change in the sequence of nucleotides of DNA or gene is called:

(a) Variation

(b) Speciation

(c) Mutation ✓

(d) Hybridization

(3) Differences in the same basic characters is called:

(a) Morphology

(b) Homology

(c) Analogy

(d) Variations ✓

(4) The change in an organism after many generations is due to:

(a) Diseases

(b) Evolution ✓

(c) Over production

(d) Deficiency of food

(5) When different species of dogs interbreed new species of dogs are formed. This process is called:

(a) Immunization

(b) Vaccination

(c) Hybridization ✓

(d) Evalution

(6) Different eye colours in human beings is an example of:

(a) Evolution

(b) Variation ✓

(c) Speciation

(d) Hybridization

(7) The direct evidence of evolution is:

(a) Embryology

(b) Palaeontology ✓

(c) Physiology

(d) Homology

(8) Analogous organs have:

(a) Same function and same structure

(b) Different function and same structure

(c) Same function and different structure ✓

(d) Different function and different structure

(9) The process through which new species are formed from a common ancestor is called:

(a) Hybridization

(b) Variation

(c) Interbreeding

(d) Speciation ✓

B. Short response questions.

(i) List down the observations of Darwin which he noted at Galapagos islands.

On the Galapagos islands Darwin observed 13 species of finches, each with a different beak shape suited to its food, giant tortoises differing from island to island, and animals found nowhere else. He noted that species varied with place and seemed adapted to their own island environment.

(ii) How was Darwin impressed by the evidence obtained from selective breeding?

Darwin was impressed that breeders, by selecting parents with desired characters over generations, could produce very different pigeons, dogs and crops from common ancestors in a short time. He reasoned that if human selection could do so much, natural selection over long ages could produce new species.

(iii) What are vestigial organs? Give some examples of vestigial organs found in human beings.

Vestigial organs are organs that were useful in ancestors but have lost most or all of their function in the present organism. In human beings examples are wisdom teeth, ear muscles, the tail bone (coccyx) and the nictitating membrane of the eye.

(iv) Define the following terms: (i) Mutation (ii) Evolution (iii) Natural selection (iv) Species (v) Anatomy (vi) Palaeontology.

(i) Mutation: a sudden change in the nucleotide sequence of DNA or a gene. (ii) Evolution: slow, gradual change in living things over a long span of time. (iii) Natural selection: survival and reproduction of the best-adapted individuals, so their characters spread. (iv) Species: a group of similar organisms that can interbreed to produce fertile offspring. (v) Anatomy: the study of the internal structures of organisms. (vi) Palaeontology: the study of

fossils, the remains of past organisms buried in soil or rocks.

C. Extended response questions.

(i) Explain the theory of evolution by natural selection in detail.

Darwin and Wallace proposed that organisms produce more offspring than can survive, so there is a struggle for existence. Individuals vary, and those with characters better suited to the environment survive and reproduce more (natural selection). Over many generations these useful characters accumulate, populations change, and new species arise. The theory is supported by fossils, comparative anatomy, embryology and selective breeding.

(ii) Write a comprehensive note on those sources of variation that can lead to speciation.

Sources of variation are mutation (sudden changes in DNA), recombination and crossing over during meiosis (new gene combinations), independent assortment of chromosomes, and hybridisation between different varieties. When a population is separated (for example by geography) and accumulates different variations under different selection pressures, it can no longer interbreed with the original population, and a new species forms (speciation).

(iii) Describe the differences between homology and analogy with examples.

Homologous organs have the same fundamental structure and origin but are modified for different functions, for example the forelimbs of a human, a bat and a whale, all built on the same bone plan. Analogous organs have the same function but different structures and origins, for example the wings of a bird and of an insect, both used for flight but built differently. Homology shows common ancestry; analogy shows similar adaptation.